P-S-411a
May 1, 1967
SUPERSEDING
Fed. Spec. P-S-411
April 20, 1955

FEDERAL SPECIFICATION

SKIN PROTECTIVE COMPOUND, CHEMICAL BARRIER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers three types of skin protective compounds.

1.2 Classification.

1.2.1 Types. This specification covers skin protective compounds of the following types:

I—Water dispersible ointment.
II—Water miscible ointment.
III—Water insoluble ointment.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

O-T-236—Tetrachloroethylene (Perchloroethylene); Technical Grade.
UU-P-150—Paper, Carbon, Duplicating Machine, Spirit-Process.
VV-B-231—Benzene (Benzol); Technical.
VV-C-846—Cutting Fluids: Emulsifiable Oils.
VV-C-850—Cutting Oil, Sulfurized-Fatty-Mineral.
VV-G-76—Gasoline, Automotive.
VV-O-251—Oil; Cutting, Mineral-Fatty-Oil-Blend.
PPP-B-636—Box, Fiberboard.
PPP-C-186—Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 6850

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:17Z
Check the source to verify that this is the current version before use.

Military Specification:

MIL-C-15203—Coating Compound, Bituminous, Emulsion Type, Coal Tar Base.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Federal Food, Drug, and Cosmetic Act, and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

3. REQUIREMENTS

3.1 Material. The ingredients used in the manufacture of skin protective compounds furnished under this specification shall conform to the standards set forth in the Federal Food, Drug, and Cosmetic Act.

3.2 Homogeneity and freedom from grit. The product shall be homogeneous, soft in texture, and free from grit, when tested as specified in 4.3.6.

3.3 Consistency. The product shall be of such consistency that at temperatures of 40° to 100° F. it will apply readily to the skin to form a flexible adhering film which shall not be sticky, stiff, nor greasy, when determined as specified in 4.3.6.

3.4 Removability. When tested as specified in 4.3.6, the product shall be readily removable from the skin with soap and warm water.

3.5 Odor. When tested as specified in 4.3.6, the product shall present no objectionable odor during and after application to the skin.

3.6 Stability. The product shall be stable to storage conditions at temperatures from minus 60° F. (slow freezing) to plus 125° F., when tested as specified in 4.3.6.

3.7 Barrier protection.

3.7.1 U. S. Public Health Service method. When tested as specified in 4.3.6, the product shall show no penetration, before 30 minutes have elapsed, to the specific challenging agents of table I.

TABLE I. Specific challenging agents

Type	Challenging material	Specification number	Indicator	Color
I	Coating compound, bituminous emulsion	MIL-C-15203	None required	Black or brown
II	Gasoline, automotive	VV-G-76	None required	Color of challenging fluid
	Tetrachloroethylene (perchloroethylene)	O-T-236	Sudan red	Red
	Benzene (benzol); technical	VV-B-231	Sudan red	Red
	Cutting oil, soluble	VV-C-846	Sudan red	Red
	Cutting oil, sulfurized-fatty-mineral	VV-C-850	Sudan red	Red
	Oil, cutting, mineral-fatty-oil-blend	VV-O-251	Sudan red	Red
III	Hydrochloric acid, C. P. (0.25N)		Bromphenol blue	Yellow
	Sodium hydroxide C. P. (0.25N)		Phenolphthalein	Red

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

3.7.2 Use test. Type I skin protective compound shall conform satisfactorily when subjected to the use test in 4.3.6.

3.8 pH value. When tested as specified in 4.3.6, the pH shall fall within the limits of 6.0 and 8.5.

3.9 Toxicity. The skin protective compound shall conform to the Federal Food, Drug, and Cosmetic Act concerning regulations on cosmetics.

3.10 Labeling. Each container of skin protective compound shall be clearly and legibly labeled. The label shall be a light color, printed on opaque paper stock in black with an ink which is legible after immersion in water at 70° to 80°F. for one hour. Labels shall be securely affixed in place with water-resistant label adhesive as specified in section 3 of MIL-STD-129 and shall be waterproofed by coating the entire outer surface of the label with the same adhesive. The label data shall be as follows:

SKIN, PROTECTIVE COMPOUND
Type
For protection against—(as given in appropriate portion of 6.1)
Contents

Warning: Do not depend on use of this product for absolute protection. Following an accidental exposure of the skin to harmful chemicals, the skin should be thoroughly washed and the protective compound should be reapplied. Wash skin thoroughly with soap and water before eating or leaving the job. Do NOT use this product for protection against any material not named above without first consulting your foreman.

DIRECTIONS FOR USE

To apply or reapply compound, wash and dry skin thoroughly before applying compound, then rub on enough compound to cover skin with a thin even film. Let film dry.

To remove compound, thoroughly wash compound coated areas of skin with soap and warm water at the prescribed hours during and at the end of your work shift.

Report promptly to the dispensary whenever you have any skin irritation.

3.11 Workmanship. The skin protective compound shall be a homogenious preparation, free from grit, dirt or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.5. The lot size shall be expressed in filled unit containers for examination in 4.3.2 and 4.3.3 and in units of shipping containers for examination in 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:17Z
Check the source to verify that this is the currentversion before use.

4.3.2 Examination of unit containers. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Container	Not type specified.
	Closure not as specified.
Contents	Not homogeneous and soft in texture.
Marking	Not in accordance with contract and specification requirements.
	Illegible, incorrect, or incomplete.
	Unit container not labeled as specified (see 3.10).

4.3.3 Examination of the filled unit containers for compliance with net contents. The end item shall be examined for defects in net contents. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container examined is less than the quantity specified.

4.3.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking requirements comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. The lot shall be the number of shipping containers submitted for inspection at one time. Closed shipping containers shall be examined for closure defects.

Examine	Defect
Markings	Incorrect; illegible; omitted; of improper size, location, sequence, or method of application (see 5.3).
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping, or stapling.
	Bulged or distorted container.
Count	Number of intermediate packages not as specified.

4.3.5 Inspection levels and AQLs for examinations. The inspection levels and AQLs expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.2	I	2.5
4.3.3	S-2	N.A.
4.3.4	S-1	2.5

4.3.6 Testing of the end item. The methods of testing listed in table II shall be followed for each lot. For sampling purposes, the lot shall consist of the finished skin protective compound of one type manufactured as one batch. The sample unit shall be 3 one pound containers selected at random throughout the lot in the sample size indicated below. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

Lot size (pounds)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

TABLE II. Instructions for testing of the end item

Characteristic	Specification reference		Requirements applicable to		Number determinations per unit	Results reported as		Inspect, level	AQL
	Requirement	Test method	Individ. unit	Lot aver.		Pass or fail	Numerically to nearest		
Material	[1] 3.1								
Homogeneity and freedom from grit	3.2	4.4.1	X		1	X			
Consistency	3.3	4.4.2	X		1	X			
Removability	3.4	4.4.3	X		1	X			
Odor	3.5	4.4.4	X		1	X			
Stability	3.6	4.4.5	X		1	X			
Barrier protection	3.7	4.4.6	X		1	X			
pH value	3.8	4.4.7	X		1	X			
Toxicity	[1] 3.9								

[1] Supplier shall furnish a certificate of compliance covering this requirement.

4.4 Test procedures.

4.4.1 Freedom from grit. Spread approximately 1 ml. of the product to a thin, even film on a glass plate. Examine to determine the presence or absence of grit.

4.4.2 Consistency.

4.4.2.1 Cold application. Cool product in closed container in an atmosphere maintained at 40° ± 2°F. for five hours. In air at this temperature and relative humidity at 80 ± 5 percent, apply product to fingers, hands, and forearms as directed in 3.10. Note ease of application, smoothness and continuity of film on skin. After five minutes, flex fingers for one minute, then examine to note whether film has cracked off, is sticky, or greasy.

4.4.2.2 Warm application. Warm product in an atmosphere maintained at 100° ± 5°F. for five hours. In air at this temperature and at a relative humidity of 30 ± 5 percent, apply small quantity of product to fingers, hands and forearms as directed in 3.10. Note ease of application, smoothness and continuity of film on skin. Flex fingers for one minute, then examine to note whether film has cracked off, is sticky, or greasy.

4.4.3 Removability. After tests specified in 4.4.2.1 and 4.4.2.2 above, wash hands and arms with warm water and a good grade of soap. The ointment shall be readily removed without the need for the abrasive action of a cloth or brush.

4.4.4 Odor. Note odor of product in freshly opened container, during application to the skin, and five minutes after cream has been applied to skin. The product shall not leave an unpleasant odor before or while performing the test in accordance with 4.4.2.1 and 4.4.2.2.

4.4.5 Stability.

4.4.5.1 Low temperature. Place two unopened containers in a refrigerated space maintained at minus 60° ± 5°F. After twenty-four hours, remove samples from refrigerator, allow samples to reach room temperature (60° to 90°F.), repeat cooling and warming cycle once more, then open one container and examine contents for separation or nonhomogeneity. Unopened sample shall be used in high temperature test in accordance with 4.4.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

4.4.5.2 High temperature. Place unopened container with sample from cold test in an oven maintained at 125° ± 2°F. After seven days at 125°F., remove container from oven and allow to cool to room temperature. Open container and examine contents for separation, nonhomogeneity, and unpleasant odor. The sample shall meet both low and high temperature stability exposures before unexposed samples are subjected to the barrier and use tests.

4.4.6 Protection or barrier efficiency. The protection or barrier efficiency tests shall be conducted at a temperature of 77° ± 2°F. and 55 ± 5 percent relative humidity. Using separate samples, the barrier efficiency shall be tested immediately after the film is applied and one hour after the film is applied. The challenging materials for types I, II, and III ointments shall conform respectively to materials listed in table I.

4.4.6.1 U. S. Public Health Service method. The following technique (see 6.3), shall be used for the evaluation of types I, II, and III ointments: Impregnate one piece of the confined spot test paper [1] with the proper indicating solution listed in table I, and allow to dry before using in the test assembly. The ointment to be tested is then loaded into a calibrated glass syringe, care being taken to avoid the introduction of air bubbles. Press 0.1 cc. of the ointment onto the center of the impregnated test paper. Place an open sheet of brass shim 0.005 inch (0.127 mm.) thick on top of the paper and around the ointment; on top of this place another piece of confined spot test paper, the confined area being carefully centered on the confined area of the bottom sheet (see fig. 1). Subject the whole assembly to 5,000 pounds pressure (see 6.4) so that the maximum thickness of the ointment film becomes that of the shim metal standard.[2] Carefully transfer the ointment "sandwich" to a viewing box (fig. 2) and then place approximately 0.1 cc. of the proper challenging agent (table I) on top of the assembly (see fig. 1). The volume of challenging solution used shall be in excess of that absorbed by the paper. By means of a mirror, illuminate the bottom of the test assembly with a fluorescent laboratory light. Penetration is the appearance of color in the bottom paper, care being taken to distinguish between color due to transmitted light and that due to a true breaking of the ointment film. Any penetration of the specific challenging agents, before 30 minutes have elapsed, indicates a failure to meet this requirement.

4.4.6.2 Use test (type I only). Thoroughly wash hands with soap and water, then dry hands thoroughly. Apply coating of protective compound to the fingers and hands as recommended in 3.10. Place one hand with palm uppermost on a flat horizontal surface. Place a fresh sheet of paper; duplicator, liquid process, conforming to UU-P-150 with dye face in contact with palm of the hand. Then place a stiff flat surface such as a board or stiff cover book on top of paper with weights so that total weight on the hand is five pounds. Revolve paper around a vertical axis passing approximately through the midpoint of palm, until five revolutions have been completed. Remove paper and weight from hand. After one hour, wash soiled hand with water. There shall be no significant difference in color of skin of both hands.

4.4.7 pH determination. A five g. sample of the compound shall be heated to boiling with 45 ml. of boiled distilled water, then allowed to cool to room temperature. Using a glass electrode known to have a low sodium error, determine electrometrically pH of water solution.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. One pound of the skin protective compound shall be packaged in a glass jar conforming to group A, class 1, type D, style 4, grade 2, closure A of PPP-C-186.

5.1.1.2 Intermediate packaging. Twelve jars of one type of skin protective compound, as specified in 5.1.1.1, shall be intermediately packaged in accordance with procedure code No. 6 of the intermediate packaging requirements of PPP-C-186.

---

[1] The confined spot test paper is a small square of filter paper at the center of which is set off or "confined" a circular area of approximately 1 square centimeter by means of a ring of paraffin with which the paper is impregnated. A drop of solution placed on the central area thus remains confined to an area of 1 square centimeter.

[2] The preparation of the standard shim is a matter of importance. The metal is so thin that in cutting it or forming it with punches the edges tend to be badly turned. This may be remedied by careful smoothing with fine, flat files.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. The skin protective compounds shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Forty-eight pounds of one type of skin protective compound, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class weather-resistant grade V3c or V3s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing by means of flat steel strapping or tape banding shall be in accordance with the appendix of the container specification.

5.2.2 Level B. Unless otherwise specified (see 5.2.2.1), 48 pounds of one type of skin protective compound shall be packed in a snug-fitting fiberboard shipping container conforming to class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of the container specification.

5.2.2.1 When specified (see 6.2), weather-resistant fiberboard specified in 5.2.1, shall be used for the shipping container.

5.2.3 Level C. The skin protective compounds, packaged as specified in 5.1.1, shall be packed to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Skin protective compounds covered by this specification are intended for use on human skin to assist in protecting the wearer against skin contact with the following:

Type I—Staining and adhering compounds.
Type II—Hydrocarbons, oils, solvents, etc.
Type III—Dilute acids, alkalies, and other water miscible chemicals.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) The type required (see 1.2.1).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Whether weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).

6.3 The technique and equipment in the U. S. Public Health Service method are those described by Louis Schwartz, Howard S. Mason, and Heloise R. Albritton in "A Method for the Evaluation of Protective Ointments", from Occupational Medicine, April 1946, Vol. 1, pp. 376 to 385.

6.4 Satisfactory results have been obtained by utilizing a Carver laboratory press.

Source: https://assist.dla.mil -- Downloaded: 20260-26T11:17Z
Check the source to verify that this is the current version before use.

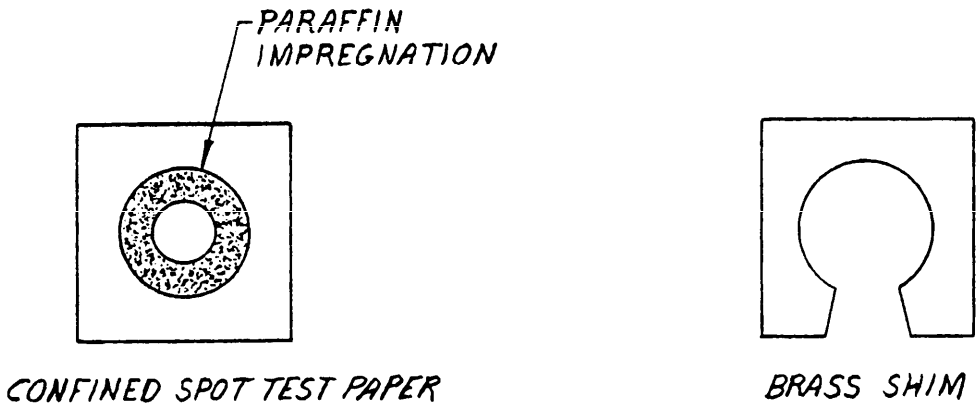
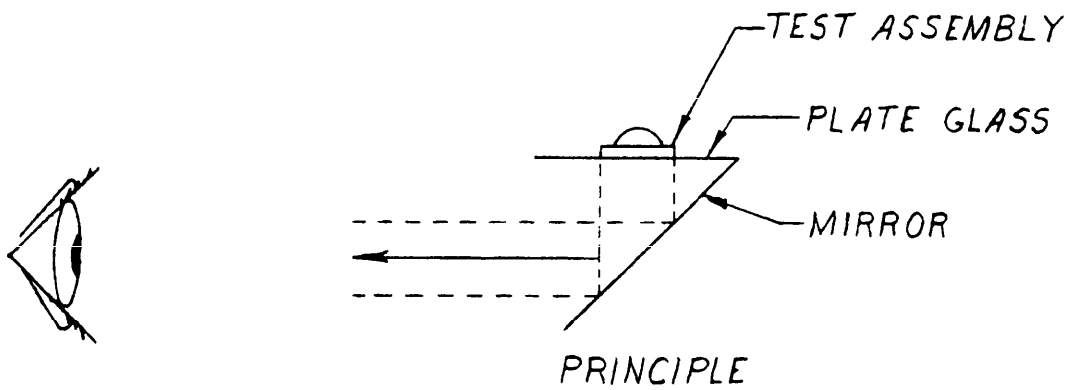


Figure 1. Ointment test assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.



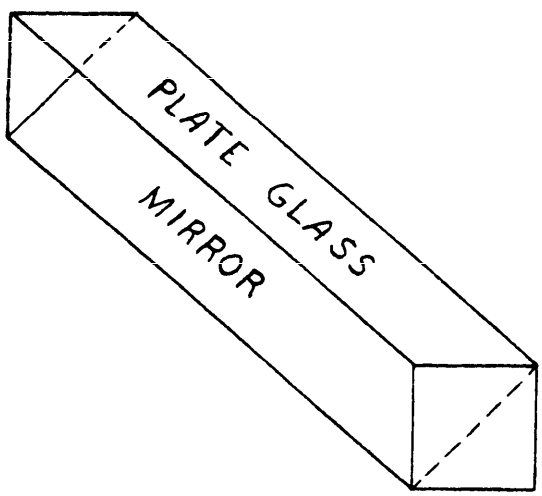


Figure 2. Viewing Box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

MILITARY INTEREST:

Custodians:
Army—GL
Navy—MS
Air Force—68

User activities:
Army—GL, MD, MU
Navy—MS, CG, MC
Air Force—68, 11

Preparing activity:
Army—GL

CIVIL AGENCY INTEREST:

Interested activities:
AGR
COM
GSA
HEW
INT
JUS
VA

Review and user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Skin Protective Compound, Chemical Barrier - P-S-411a

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01762

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRE-SS
Natick, Massachusetts 01762

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:17Z
Check the source to verify that this is the current version before use.